EXPERIMENT – 1

Aim- To design an Entity-Relationship (ER) diagram for an Indian E-commerce platform consisting of Customer, Product, Order, Order Item, Seller, Category, Payment, Delivery and Address entities, while identifying primary keys, composite attributes, multi-valued attributes, weak entities, specialization and participation constraints.

Objective-

- To identify the entities involved in an Indian e-commerce system.
- To identify the attributes of each entity.
- To identify primary keys for the entities.
- To identify composite and multi-valued attributes.
- To identify a weak entity and its identifying relationship.
- To represent relationships between entities.
- To represent cardinality and participation constraints.
- To represent specialization of the Product entity.

Theory

1. Entity

An entity is a real-world object about which information is stored.
For our e-commerce system, examples are:

- Customer
- Product
- Order
- Seller
- Category
- Payment
- Delivery
- Address

2. Entity and Primary Key

A **primary key** uniquely identifies each occurrence of an entity.
For example:

- Customer
- CustomerID ← Primary Key
- CustomerName
- Email

3. Composite Attribute

A composite attribute can be divided into smaller attributes.
For example: Address can be divided into Street, City, State, Pincode

4. Multi-valued Attribute

A multi-valued attribute can have **more than one value** for one entity.
For example, a customer can have multiple phone numbers

5. Weak Entity

A **weak entity** does not have a complete primary key of its own for unique identification.
For this experiment, we treat **Address** as a weak entity.

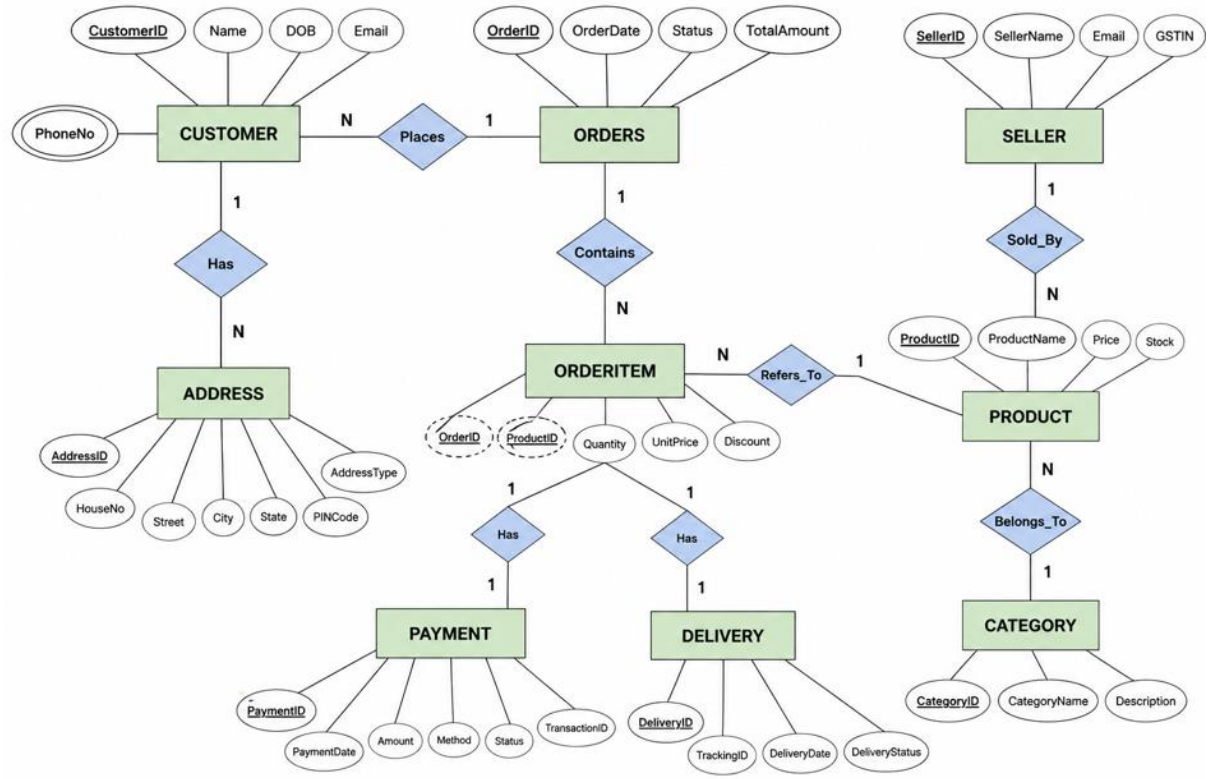
6. Composite Primary Key

A composite primary key contains more than one attribute.

7.Participation Constraints

Participation tells us whether participation in a relationship is **mandatory or optional**.

ER Diagram:



Output:

1. The ER diagram for the Indian E-commerce Platform was successfully designed.
2. The entities Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery and Address were represented.
3. Primary keys, composite keys and multi-valued attributes were identified in the ER model.
4. Address was represented as a weak entity, and Product was specialized into Electronics and Clothing.
5. The required relationships, cardinalities and participation constraints were represented using the ER diagram.

Observation/Outcomes:

1. The entities and their attributes were identified according to the requirements of the e-commerce system.
2. Relationships between customers, orders, products, sellers, categories, payments, deliveries and addresses were established.

3. The ER model successfully represents the required weak entity, composite attribute, multi-valued attribute and specialization.
4. Primary keys and relationship cardinalities help define how the entities are connected.
5. Thus, the ER diagram provides a clear conceptual representation of the Indian E-commerce database.